

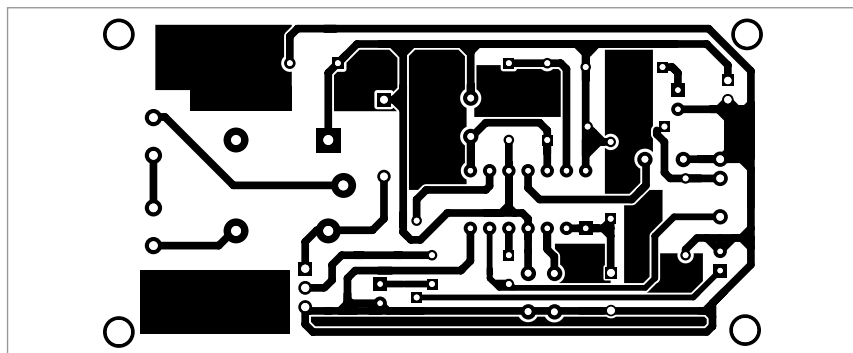


Fig. 5: Actual-size PCB layout

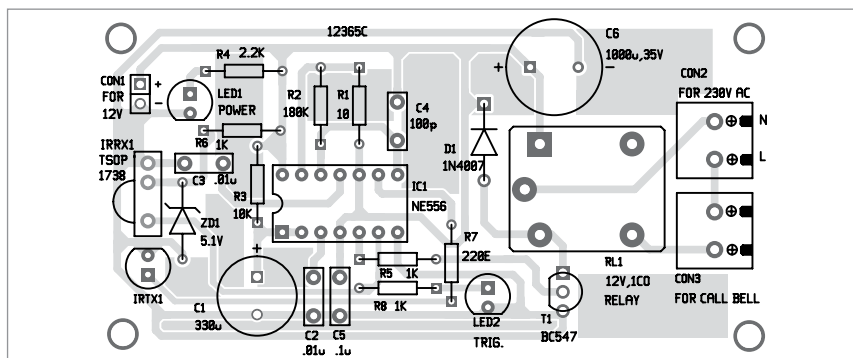


Fig. 6: Components layout of the PCB



Fig. 7: Internal view of the author's prototype

sensor. The astable mode is configured with resistors R1 and R2 and capacitor C4. The astable frequency calculation can be done using the relationship:

$$F = 1.44 / (R1 + 2 \times R2) \times C4$$

The frequency output obtained at pin 9 of IC1 is connected to IRTX1 LED through resistor R7, which controls the current flowing through the LED. The IR light emitted is proportional to this current. The value of R7 can be changed to adjust the proximity range of the system. The forward voltage of IRTX1 is around 1.9V. Zener diode ZD1 connected in series with resistor R6 is used to generate 5.1V DC, which drives the sensor.

In the monostable multivibra-

tor section (pins 1 through 6), resistor R3 and capacitor C1 provide the necessary timing. Here the timing (on state) is around three seconds. The timing calculation is given by the relationship:

$$T = 1.1 \times R3 \times C1$$

When reflected IR beam is received by the sensor, its output pin 3 becomes low. This output goes to trigger pin 6 of IC1 (monostable multivibrator section). The monostable output becomes high when a low signal is received at its trigger pin 6. The output pin 5 of IC1 is connected to the base of transistor T1 through resistor R5. Transistor T1 conducts when output pin 5 of IC1 goes high.

Relay RL1 connected to collector of transistor T1 energises when the transistor conducts. Thus, transistor T1 switches the relay on/off depending on the signal received. The doorbell is connected across the relay contacts as shown in the circuit diagram. Doorbell rings when the relay energises.

PARTS LIST

Semiconductors:

- IC1 - NE556 dual timer
- T1 - BC547 npn transistor
- D1 - 1N4007 rectifier diode
- LED1, LED2 - 5mm LED
- ZD1 - 5.1V Zener diode
- IRTX1 - IR transmitter LED
- IRRX1 - TSOP1738 module (IR receiver)

Resistors (all 1/4-watt, ±5% carbon):

- R1, R3 - 10-kilo-ohm
- R2 - 180-kilo-ohm
- R4 - 2.2-kilo-ohm
- R5, R6, R8 - 1-kilo-ohm
- R7 - 220-ohm

Capacitors:

- C1 - 330μF, 25V electrolytic
- C2, C3 - 0.01μF ceramic disk
- C4 - 100pF ceramic disk
- C5 - 0.1μF ceramic disk
- C6 - 1000μF, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- CON2, CON3 - 2-pin terminal connector
- RL1 - 12V, single-changeover relay
- Jumper wires
- 230V AC doorbell
- 8-pin socket for IC1

LED1 is used as a power on indicator and LED2 connected to pin 6 of IC1 through R8 is used as output signal indicator.

Construction and testing

An actual-size PCB layout for the doorbell is shown in Fig. 5 and its components layout in Fig. 6. After assembling the circuit, connect the 12V across CON1, 230V AC mains across CON2, and 230V AC doorbell across CON3.

Enclose the circuit in a suitable box. Fix IRTX1 and IRRX1 sensors near each other facing towards the user, as shown in the prototype. When you keep your finger in front of these sensors, the transmitted IR beam is received by sensor TSOP1738 to trigger the doorbell.

Before assembling and installing the doorbell, you may refer to author's assembled circuit shown in Fig. 7. On the left side is 12V SMPS circuit and on right side is the main doorbell circuit. Both the circuits are enclosed in a plastic cabinet. **EFY**



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